Xinzhi (Berry) Liu, PhD, MIEEE

Postdoctoral Research Fellow in Power Electronics | GaN and SiC Device Reliability

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RESEARCH PROFILE

I study the reliability and robustness of GaN and SiC power devices at circuit level. I design converter-level test platforms that separate electrical stress history from the measurement pulse, so that trapping, thermal and switching effects can be told apart in commercial devices, and I use SPICE and TCAD to connect the measurements to device physics. My current work covers reverse conduction, current collapse and short-circuit ruggedness of 650 V monolithic bidirectional GaN switches.

Research interests: GaN HEMT and bidirectional switch reliability (dynamic $R_{DS(on)}$, current collapse, V_{TH} instability, third-quadrant conduction); short-circuit, UIS, surge and linear-mode robustness of wide bandgap devices; converter-level test methodology and measurement circuits; electrothermal characterisation and thermal impedance; SPICE and TCAD modelling; wide bandgap hybrid switches.

Record at a glance (November 2026): 27 peer-reviewed publications (19 journal, 8 conference), 12 as first or corresponding author; 5 IEEE journal papers, including first-author articles in *IEEE Transactions on Power Electronics* and *IEEE Transactions on Device and Materials Reliability*; Google Scholar: 90 citations, h-index 5; 97 verified peer reviews for 13 journals.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Postdoctoral Research Fellow in Power Electronics **Sep 2025 to Present**
PERC Lab, School of Engineering, University of Warwick, UK. PI: Prof. Layi Alatise; with Dr Jose Ortiz Gonzalez

- Research Fellow on the UKRI project *Circular Microgrids: Circular Economy Pathways for Renewable Microgrids in Africa* (UKRI-320), working on power semiconductor reliability.
- Conceived and built a converter-level third-quadrant test platform (full bridge with an isolated pre-charged inductor) that tests 40 V to 650 V GaN devices free of prior off-state voltage stress. Reverse-conduction voltage measured 8% to 35% above datasheet values; trapping inflates apparent thermal impedance by up to 7 times (ECCE Europe 2026).
- Developed an isolated stress-dwell technique for stress-history-dependent current collapse in bidirectional GaN HEMTs (*IEEE TDMR*, 2026), now the GaN current-collapse function of the group's integrated robustness test system (*IEEE JESTPE*, 2026).
- Lead test-platform development for the 650 V dual-gate GaN bidirectional switch programme: multi-board adapter with dual isolated, Kelvin-referenced gate drivers for short-circuit, UIS and surge testing, and a repetitive ZVS pulse platform with selectable gate states.
- Contribute to group studies on 1,200 V SiC linear-mode robustness, SiC short-circuit and avalanche ruggedness (with GE Vernova) and 1,200 V GaN/IGBT and SiC/IGBT hybrid switches.

Visiting PhD Researcher (SUSTech International Fellowship) **Sep 2023 to Mar 2025**
Department of Electrical and Electronic Engineering, Southern University of Science and Technology (SUSTech), Shenzhen, China. Host: Prof. Mengyuan Hua

- Designed four-layer, low-inductance GaN converter PCBs with top-side cooling and a multi-pulse platform for dynamic $R_{DS(on)}$ and C_{OSS} extraction under high-speed switching, leading to a first-author *IEEE Transactions on Power Electronics* paper.

Technical Support (Substation Design) **Apr 2019 to Feb 2020**
Manitoba Hydro, Manitoba, Canada

- Protection-system upgrade design for the Raven Lake and Virden West substations, a 4 kV transformer decommissioning plan and commissioning records.

EDUCATION

2025 **PhD, Electrical Power Engineering**, Universiti Putra Malaysia (Sep 2022 to Sep 2025; viva May 2025). Supervisor: Prof. Suhaidi Shafie. Thesis: *Dynamic On-Resistance Characterization and Multi-Pulse Switching Output Capacitance Extraction of Stabilized Schottky p-GaN HEMT for Power Converter Applications*. Ranked 1st in Faculty (GPA 4.0/4.0). Included an 18-month funded research attachment at SUSTech, China.

2022 **MSc Green Technology (Energy), Distinction**, Hong Kong Baptist University, Hong Kong SAR (Aug 2021 to Nov 2022). Ranked 1st in programme (GPA 3.8/4.0). Research on nanostructured electron transport layers for perovskite solar cells.

2020 **Diploma (Honours), Electrical Engineering Technology**, Red River College Polytechnic, Winnipeg, Canada (Apr 2017 to Feb 2020). GPA 4.31/4.5.

FELLOWSHIPS, AWARDS AND FUNDING

2025 Best PhD Dissertation Award 2025, IEEE Circuits and Systems Society (CASS) Malaysia Chapter.

2025 IEEE Power Electronics Society (PELS) travel award, IDEAS 2025 (USD 500).

2025 to 2027 Research Fellow (from Sep 2025) on *Circular Microgrids* (UKRI-320), EPSRC/UKRI International Science Partnerships Fund, Ayrton Challenge Programme; £2.7M multi-institution award led by De Montfort University. Employed on the award, not an investigator.

2023 to 2024 SUSTech International Fellowship for visiting PhD research, Shenzhen (RMB 200,000, personal award).

2022 to 2025 UPM Graduate Student Scholarship (PhD).

2022 Scholastic Award, Hong Kong Baptist University.

2018 National Award (team), Electric Mobility Canada Kia Student Competition.

PUBLICATIONS

All items are peer reviewed. Own name in bold. J = journal article, C = conference paper. Notes under selected papers state my contribution.

A. Wide bandgap power devices and power electronics

Journal articles, first author

[J1] **Liu, X.**, Ayala, H., Mahmood, A., Zhang, M., Ortiz Gonzalez, J. and Alatise, O. (2026). A stress-dwell technique for current-collapse evaluation in bidirectional normally off p-GaN HEMTs. *IEEE Transactions on Device and Materials Reliability*, early access. doi:10.1109/TDMR.2026.3751157.
Conceived the technique; designed the isolated stress, dwell and pulse platform; performed all measurements and analysis; wrote the manuscript. Off-state stress of 0 to 20 V raises third-quadrant conduction voltage by up to 28% and conduction power by up to 42% in a 40 V bidirectional GaN HEMT.

[J2] **Liu, X.**, Chen, J., Wang, S., Wu, S., Jiang, Z., Radzi, M. A. M., Azis, N., Shafie, S. and Hua, M. (2026). Output capacitance extraction of p-GaN HEMTs under multi-pulse switching across variable frequencies and drain-bias stress. *IEEE Transactions on Power Electronics*, 41(6), pp. 8927-8937. doi:10.1109/TPEL.2026.3651976.
Conceived the extraction method; designed the four-layer low-inductance PCB and top-cooled heat sink; built the multi-pulse platform; performed all measurements, analysis and writing. Separates OFF-state electron trapping from reverse-conduction hole injection in the measured C_{OSS}.

[J3] **Liu, X.**, Shafie, S., Radzi, M. A. M., Azis, N., Abdul Karim, A. H., Norddin, N. and Lawal, I. (2025). Simulation-based evaluation of power efficiency and output capacitance in standalone PV MPPT buck converters using 200 V p-GaN HEMTs. *Journal of Electrical Engineering & Technology*, 20, pp. 3875-3887. doi:10.1007/s42835-025-02334-y.

[J4] **Liu, X.**, Hua, M., Shafie, S., Radzi, M. A. M. and Azis, N. (2024). SPICE modelling-assisted evaluation of dynamic on-resistance characterization in Schottky p-GaN HEMTs amid synchronous buck transient instabilities. *Computers & Electrical Engineering*, 118, 109410. doi:10.1016/j.compeleceng.2024.109410.
Introduces a SPICE-assisted method to separate the intrinsic dynamic $R_{DS(on)}$ of Schottky p-GaN HEMTs from converter transient artefacts.

[J5] **Liu, X.**, Shafie, S., Radzi, M. A. M., Azis, N. and Abdul Karim, A. H. (2024). Modelling and mitigating oscillation in E-mode GaN HEMT: a simulation-based approach to parasitic inductance optimization. *Microelectronics Reliability*, 152, 115293. doi:10.1016/j.microrel.2023.115293.

[J6] **Liu, X.**, Shafie, S., Radzi, M. A. M., Azis, N., Norddin, N., Lawal, I., Zulkifli, N. and Abdul Karim, A. H. (2024). SPICE modelling and performance analysis of enhancement-mode GaN HEMTs for augmented hard-switching energy conversion efficiency. *Pertanika Journal of Science & Technology*, 32(3), pp. 1243-1262. doi:10.47836/pjst.32.3.14.

Journal articles, co-author

[J7] Alatise, O., Ayala, H., **Liu, X.**, Deb, A., Mahmood, A., Jahdi, S. and Ortiz Gonzalez, J. (2026). Integrated robustness test systems of wide bandgap power devices. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, early access. doi:10.1109/JESTPE.2026.3751357.
Contributed the GaN current-collapse evaluation (Section VII): adaptation of the test rig, measurements and text.

[J8] Mahmood, A., Ayala, H., Wan, H., **Liu, X.**, Ortiz Gonzalez, J., Jahdi, S., Yuan, X. and Alatise, O. (2026). Linear mode robustness analysis of 1200 V silicon and SiC MOSFETs. *IEEE Transactions on Power Electronics*, early access. doi:10.1109/TPEL.2026.3715936.
Experimental set-up and instrumentation of the linear-mode test circuit; measurement methodology support to the PhD first author.

[J9] Ayala, H., Ortiz Gonzalez, J., Deb, A., Malik, A. H., Jahdi, S., Wan, H., **Liu, X.** and Alatise, O. (2026). Correlations between body diode pulsed overcurrent safe operating area, short circuit, and avalanche ruggedness in SiC MOSFETs. *IEEE Transactions on Electron Devices*, 73(6), pp. 3308-3320. doi:10.1109/TED.2026.3671746.
Joint paper with GE Vernova. Performed the experiments and data collection required at revision.

[J10] Abdul Karim, A. H., Ab Kadir, M. Z. A., **Liu, X.**, Azis, N., Norsahperi, N. M. H. and Abda, Z. M. K. (2026). Electromagnetic transient simulation of EV fast charging on distribution networks: comparative evaluation with PV integration. *Journal of Energy Storage*, 155, 121535. doi:10.1016/j.est.2026.121535.

[J11] Abdul Karim, A. H., Ab Kadir, M. Z. A., **Liu, X.**, Azis, N., Norsahperi, N. M. H. and Abda, Z. M. K. (2026). Harmonic distortion assessment of EV charging penetration and superconducting magnetic energy storage integration in a 400 V distribution network: EMT simulation based on a Universiti Putra Malaysia campus feeder. *Engineering Research Express*, 8, 045325. doi:10.1088/2631-8695/ae47a8.

Conference papers, first author

[C1] **Liu, X.**, Ayala, H., Ortiz Gonzalez, J., Deb, A., Mahmood, A. and Alatise, O. (2026). Evaluation of third quadrant dynamic on-resistance and thermal impedance of GaN e-HEMTs using converter level switching tests. *2026 Energy Conversion Congress & Expo Europe (ECCE Europe)*, Valencia, Spain (6 pp.). doi:10.1109/ECCEEurope67962.2026.11737951.
Joint paper with GE Vernova. Conceived and designed the test platform and device daughter boards; performed all measurements and analysis; wrote and presented the paper.

[C2] **Liu, X.**, Shafie, S., Radzi, M. A. M., Azis, N., Abdul Karim, A. H. and Norddin, N. (2025). Design of rapid-response fail-safe protection circuit for 200 V p-GaN HEMTs in LLC half-bridge converters. *2025 IEEE International Circuit and System Symposium (ICSyS)* (6 pp.). doi:10.1109/ICSYS65149.2025.11379889.

[C3] **Liu, X.**, Shafie, S., Radzi, M. A. M., Azis, N., Norddin, N., Abdul Karim, A. H. and Lawal, I. (2025). SPICE simulation assisted thermal management for surface-mount GaN HEMTs: experimental validation and performance insights. *2025 IEEE PELS International Decentralized Energy Access Symposium (IDEAS)*, pp. 1-6. doi:10.1109/IDEAS62193.2025.11157305.

[C4] **Liu, X.**, Shafie, S., Radzi, M. A. M. and Azis, N. (2024). SPICE simulation assisted dynamic $R_{DS(on)}$ characterization in 200 V commercial Schottky p-GaN HEMTs under unstable phases. *Journal of Physics: Conference Series*, 2841, 012001 (7th International Conference on Circuits, Systems and Simulation). doi:10.1088/1742-6596/2841/1/012001.

Conference papers, co-author

[C5] Zhang, M., **Liu, X.**, Ayala, H., Jahdi, S., Yuan, X. and Alatise, O. (2026). Evaluation of 1200 V wide bandgap hybrid switches: SiC MOSFET vs GaN e-HEMT/Si-IGBT. *2026 IEEE Workshop on Wide Bandgap Power Devices and Applications in Europe (WiPDA Europe)* (5 pp.). doi:10.1109/WiPDAEurope69886.2026.11746958.
Second author. PCB troubleshooting, electrothermal methodology (on-resistance as a temperature-sensitive parameter for junction temperature) and manuscript revision.

[C6] Mahmood, A., Ayala, H., Jahdi, S., **Liu, X.**, Ortiz-Gonzalez, J. and Alatise, O. (2026). Characterization of linear mode robustness of silicon and SiC power MOSFETs. *2026 Energy Conversion Congress & Expo Europe (ECCE Europe)*, Valencia, Spain (6 pp.). doi:10.1109/ECCEEurope67962.2026.11738325.

B. Photovoltaics, materials and sensing (earlier and collaborative work)

[J12] **Liu, X.** and Shafie, S. (2025). Nanostructure silver absorbance with polarized light spectroscopy (PLS) and mobile imaging. *Instrumentation Science & Technology*, 53(4), pp. 381-394. doi:10.1080/10739149.2024.2396138.

[J13] Wang, R., **Liu, X.** (corresponding author) and Shafie, S. (2024). Evaluating perovskite solar panels for thermal stability and inclination performance through finite element modelling. *Engineering Research Express*, 6, 025327. doi:10.1088/2631-8695/ad43bb.

[J14] Lawal, I., Shafie, S., Pandey, S. S., Jafaar, H., Mustafa, M. A., Abdulmalik, I. O., Ismail, I., Kamarudin, M. A., Mohd Noor, I. S., Ahmad, F. and **Liu, X.** (2025). The effect of titanium (IV) chloride surface treatment and titanium dioxide/graphenated carbon nanotube composite photoanode to enhance charge transport and light harvesting of bifacial dye-sensitized solar cell. *Solar Energy*, 294, 113495. doi:10.1016/j.solener.2025.113495.

[J15] Lawal, I., Shafie, S., Pandey, S. S., Jafaar, H., Mustafa, M. A., Ismail, I., Kamarudin, M. A., Mohd Noor, I. S., **Liu, X.**, AlSultan, H. A., Babani, S., Abdulhamid, I. B. and Norddin, N. (2025). Maximizing solar energy harvesting: enhancing the efficiency of bifacial dye-sensitized solar cells with graphenated carbon nanotubes composites in multi-layered stacked photoanode. *Optical Materials*, 160, 116721. doi:10.1016/j.optmat.2025.116721.

[J16] Zhou, F., Bi, D., **Liu, X.**, Shafie, S., Yu, X. and Qin, D. (2026). Synthesis of hollow SiO_2@C composites and their electrochemical sensing performance for quantitative detection of nitrite in food products. *ChemistrySelect*, 11, e07225. doi:10.1002/slct.202507225.

[J17] Norddin, N., Shafie, S., Idris, M. I., **Liu, X.**, Lawal, I. and Hamidon, M. N. (2026). Utilisation of response surface methodology to optimise bandgap and resistance of nickel oxide using the sol-gel method. *Journal of Engineering Technology and Applied Physics*, 8(2), pp. 32-40. doi:10.33093/jetap.2026.8.2.5.

[J18] Zhou, F., Liu, S., Zhao, Y., **Liu, X.**, Qin, D. and Liu, H. (2025). Construction of Bi-MOF/MXenes and their application in electrochemical nitrite sensors in food samples. *Journal of Food Composition and Analysis*, 148, 108511. doi:10.1016/j.jfca.2025.108511.

[J19] Norddin, N., Shafie, S., Idris, M. I., Hamidon, M. N., Ahmad, F. and **Liu, X.** (2024). Enhanced conductivity of hole transport layers via nickel oxide and graphenated-carbon nanotube (g-CNT) composites. *Malaysian Journal of Microscopy*, 20(2), pp. 186-196.

[C7] Norddin, N., Shafie, S., **Liu, X.**, Lawal, I. and Idris, M. I. (2025). Optimizing NiO properties for enhanced perovskite solar cell efficiency. *2025 IEEE International Circuit and System Symposium (ICSyS)* (5 pp.). doi:10.1109/ICSYS65149.2025.11379904.

[C8] Norddin, N., Shafie, S., Idris, M. I., **Liu, X.**, Lawal, I., Rummaja, M. I. D. M., Abdul Aziz, A. M. and Mohd Nor, R. (2023). Performance comparison of different electron transport layers for perovskite solar cell with NiO as hole transport layer using SCAPS 1D. *2023 IEEE 8th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)* (6 pp.). doi:10.1109/ICRAIE59459.2023.10468532.

C. Under review and in preparation (not counted above)

- **Liu, X.**, et al. Repetitive ZVS test platform without off-state voltage stress for 650 V GaN BDS and 150 V p-GaN. PCIM Europe 2027, digest submitted.
- Zhang, M., Ayala, H., **Liu, X.**, Mahmood, A., Wan, H., Jahdi, S., Yuan, X. and Alatise, O. Electrothermal evaluation of 1200 V GaN e-HEMT/IGBT and SiC/IGBT hybrid switches using virtual thermal impedance technique. Submitted to *IEEE Transactions on Power Electronics*.
- **Liu, X.**, et al. Short-circuit and UIS robustness of 650 V dual-gate monolithic bidirectional GaN switches. In preparation.

CONFERENCE PRESENTATIONS

Sep 2026	Oral presentation, IEEE ECCE Europe 2026, Valencia, Spain: third-quadrant dynamic on-resistance and thermal impedance of GaN e-HEMTs from converter-level switching tests.
2026	Poster, Centre for Power Electronics (CPE) Annual Conference 2026, Edinburgh, UK (University of Warwick representative).
2025	Paper presentation, IEEE PELS IDEAS 2025, supported by an IEEE PELS travel award.

SUPERVISION, MENTORING AND TEACHING

- Day-to-day technical supervision of PhD students in the PERC Lab (not the registered supervisor). Topics and outcomes: 1,200 V wide bandgap hybrid switches (WiPDA Europe 2026 paper; *IEEE TPEL* manuscript submitted);

1,200 V SiC linear-mode robustness (*IEEE TPEL* 2026 and ECCE Europe 2026 papers); HTGB gate-reliability testing (from Nov 2026).

- External mentoring of a PhD student at Universiti Putra Malaysia on research design, EMT model validation and thesis revision (*Journal of Energy Storage* 2026 and *Engineering Research Express* 2026 papers).
- Test-rig training and laboratory safety instruction for new PhD students in the PERC Lab.
- Teaching Assistant, joint dye-sensitised solar cell workshops between Universiti Putra Malaysia and Kyushu Institute of Technology, Japan (2022 and 2024; two weeks each): hands-on device fabrication, electrical testing and data analysis, and the workshop lecture on DSSC principles.

PROFESSIONAL SERVICE AND COLLABORATION

- Peer review: 97 verified reviews for 13 journals, 2023 to 2026 (ORCID record), including *IEEE Transactions on Power Electronics*, *IEEE Open Journal of Power Electronics*, *Microelectronics Reliability*, *Power Electronic Devices and Components*, *International Journal of Electrical Power & Energy Systems* and *Computers & Electrical Engineering*.
- Member, IEEE: Power Electronics, Electron Devices, Circuits and Systems, Industrial Electronics, and Power and Energy Societies.
- Industry collaboration: GE Vernova (Stafford, UK), joint papers in *IEEE Transactions on Electron Devices* and at ECCE Europe 2026.
- Academic collaboration: University of Bristol (UK); SUSTech (China); Universiti Putra Malaysia and Universiti Malaysia Sarawak (Malaysia); Kyushu Institute of Technology (Japan).

RESEARCH PLATFORMS AND TECHNICAL SKILLS

Test platforms built: converter-level third-quadrant pulse rig; isolated stress-dwell current-collapse rig; repetitive ZVS pulse platform; multi-pulse dynamic $R_{DS(on)}$ and C_{OSS} platform; dual-gate bidirectional switch adapter for short-circuit, UIS and surge testing.

Hardware: Altium Designer (multi-board, four-layer low-inductance layouts), isolated gate drivers, Kelvin sensing and on-state voltage clamp circuits, STM32 pulse-timing firmware.

Modelling: Sentaurus TCAD (device), LTspice and SIMPLIS (circuit, electro-thermal), MATLAB/Simulink and PSCAD/EMTDC (system), Ansys and COMSOL (FEM).

Instruments and automation: Keithley 2651A and 2602B SMUs, Keysight power device analysers, high-bandwidth oscilloscopes; Python, C++ and shell scripting for instrument control and data processing.

Languages: English (fluent), Mandarin (native), Cantonese and Malay (basic).

REFEREES

Available on request: current PI (University of Warwick), PhD supervisor (Universiti Putra Malaysia) and visiting-PhD host (SUSTech).